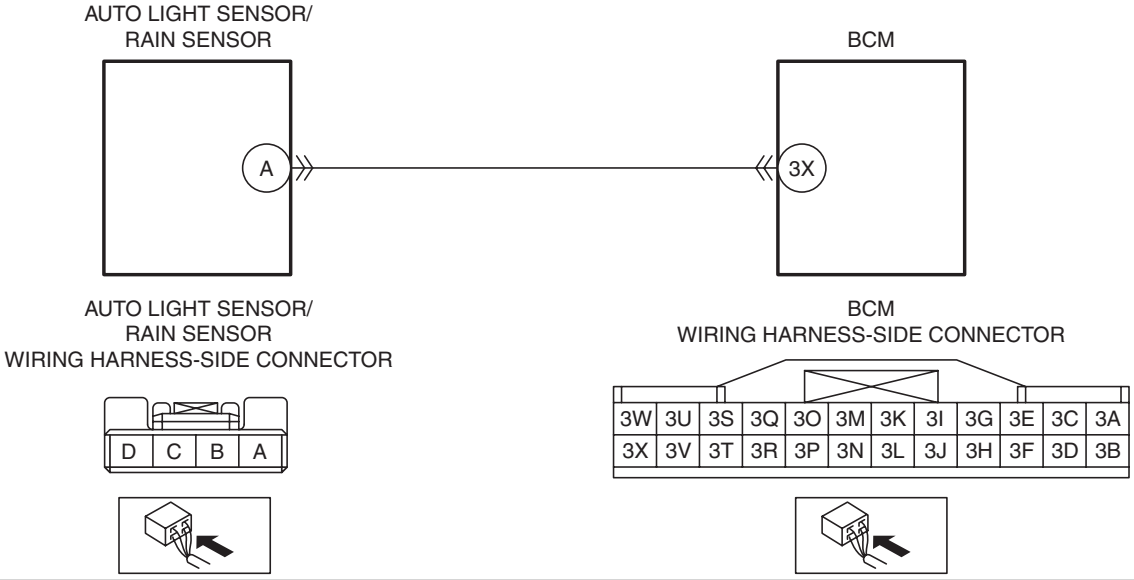
DTC U0231:68 [BCM]

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Description	Communication error with auto light sensor/rain sensor
Detection condition	• BCM could not receive the signals from the auto light sensor/rain sensor for **5 s or more** with the ignition switched ON (engine off or on).
Fail-safe	**Auto light system** • Continue the controls prior to the auto light malfunction detection. **Auto wiper system** • Continue the controls prior to the windshield wiper malfunction detection.
Possible cause	• Auto light sensor/rain sensor connector or terminal malfunction • BCM connector or terminal malfunction • Short to ground in wiring harness between auto light sensor/rain sensor terminal A and BCM terminal 3X • Short to power supply in wiring harness between auto light sensor/rain sensor terminal A and BCM terminal 3X • Open circuit in wiring harness between auto light sensor/rain sensor terminal A and BCM terminal 3X • Auto light sensor/rain sensor malfunction • BCM malfunction



Diagnostic Procedure

Step	Inspection		Action
1	**VERIFY BCM DTCs AGAIN** • Clear the DTC for the BCM using the M-MDS. (See CLEARING DTC [BCM].) • Operate the light switch to the AUTO position. • Operate the wiper and washer switch to the AUTO position. • Switch the ignition ON (engine off or on) and wait for **5 s or more**. • Retrieve the BCM DTCs using the M-MDS. (See DTC INSPECTION [BCM].) • Is the same DTC displayed?	Yes	Go to the next step.
		No	Go to Step 9.
2	**INSPECT AUTO LIGHT SENSOR/RAIN SENSOR CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the auto light sensor/rain sensor connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 8.

Step	Inspection		Action
3	**INSPECT BCM CONNECTOR CONDITION** • Disconnect the BCM connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 8.
4	**INSPECT AUTO LIGHT SENSOR/RAIN SENSOR CIRCUIT FOR SHORT TO GROUND** • Verify that the auto light sensor/rain sensor and BCM connectors are disconnected. • Inspect for continuity between BCM terminal 3X (wiring harness-side) and body ground. • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between auto light sensor/rain sensor terminal A and BCM terminal 3X. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. Go to Step 8.
		No	Go to the next step.
5	**INSPECT AUTO LIGHT SENSOR/RAIN SENSOR CIRCUIT FOR SHORT TO POWER SUPPLY** • Verify that the auto light sensor/rain sensor and BCM connectors are disconnected. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Switch the ignition ON (engine off or on). • Measure the voltage at the BCM terminal 3X (wiring harness-side). • Is the voltage **0 V**?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between auto light sensor/rain sensor terminal A and BCM terminal 3X. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to power supply. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to power supply. Go to Step 8.
6	**INSPECT AUTO LIGHT SENSOR/RAIN SENSOR CIRCUIT FOR OPEN CIRCUIT** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Verify that the auto light sensor/rain sensor and BCM connectors are disconnected. • Inspect the wiring harness for continuity between auto light sensor/rain sensor terminal A (wiring harness-side) and BCM terminal 3X (wiring harness-side). • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between auto light sensor/rain sensor terminal A and BCM terminal 3X. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 8.

Step	Inspection		Action
7	**VERIFY IF MALFUNCTIONING LOCATION IS AUTO LIGHT SENSOR/RAIN SENSOR DEPENDING ON REPEATABILITY** • Always reconnect all disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Clear the DTC for the BCM using the M-MDS. (See CLEARING DTC [BCM].) • Operate the light switch to the AUTO position. • Operate the wiper and washer switch to the AUTO position. • Switch the ignition ON (engine off or on) and wait for **5 s or more**. • Retrieve the BCM DTCs using the M-MDS. (See DTC INSPECTION [BCM].) • Is the same DTC displayed?	Yes	Replace the auto light sensor/rain sensor, then go to the next step. (See RAIN SENSOR REMOVAL/INSTALLATION.)
		No	Go to Step 9.
8	**VERIFY THAT REPAIRS HAVE BEEN COMPLETED** • Always reconnect all disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Clear the DTC for the BCM using the M-MDS. (See CLEARING DTC [BCM].) • Operate the light switch to the AUTO position. • Operate the wiper and washer switch to the AUTO position. • Switch the ignition ON (engine off or on) and wait for **5 s or more**. • Retrieve the BCM DTCs using the M-MDS. (See DTC INSPECTION [BCM].) • Is the same DTC displayed?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the BCM. (See BODY CONTROL MODULE (BCM) REMOVAL/INSTALLATION.) Go to the next step.
		No	Go to the next step.
9	**VERIFY IF OTHER DTCs DISPLAYED** • Are any other DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [BCM].)
		No	DTC troubleshooting completed.